

UHI: Urban Heat Intelligence

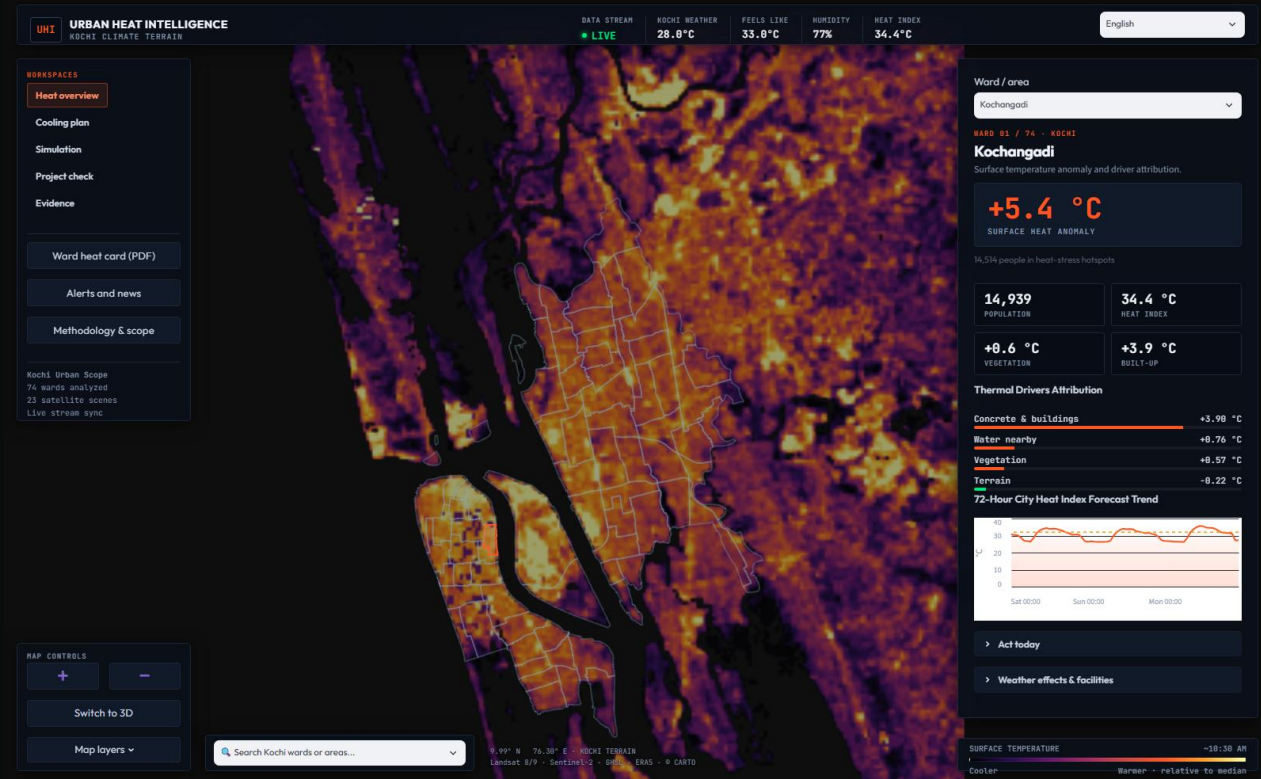
A decision-support dashboard that connects urban heat mapping to practical cooling decisions.

THE PROBLEM

Heat is uneven across the city. Planners need to know which places and people face exposure, where money can help, and whether new development adds heat.

THE PROJECT'S ANSWER

Locate hotspots → explain drivers → compare interventions → allocate budgets → screen development. Make the evidence visible at every step.



74 named wards / 54,168 study cells / 23 satellite scenes

From satellite observations to an action plan

A 100 m analytical grid combines surface conditions, weather, population and the built environment.

01 OBSERVE

Landsat 8/9 surface heat
Sentinel-2 + land cover
ERA5-Land atmosphere
GHSL population + OSM

02 EXPLAIN

Scene-panel XGBoost
Physics-informed constraints
(1 – albedo) × sunlight
TreeSHAP heat drivers

03 SIMULATE

20-neighbour local analogs
Labelled roof/pavement
formulas
Joint scenarios + spillover
Support checks and clipping

04 PRIORITISE

Person·°C benefit per rupee
Eligible public-land actions
₹1 / ₹10 / ₹50 crore plans
Types, costs and locations

EXPOSURE SCREENING

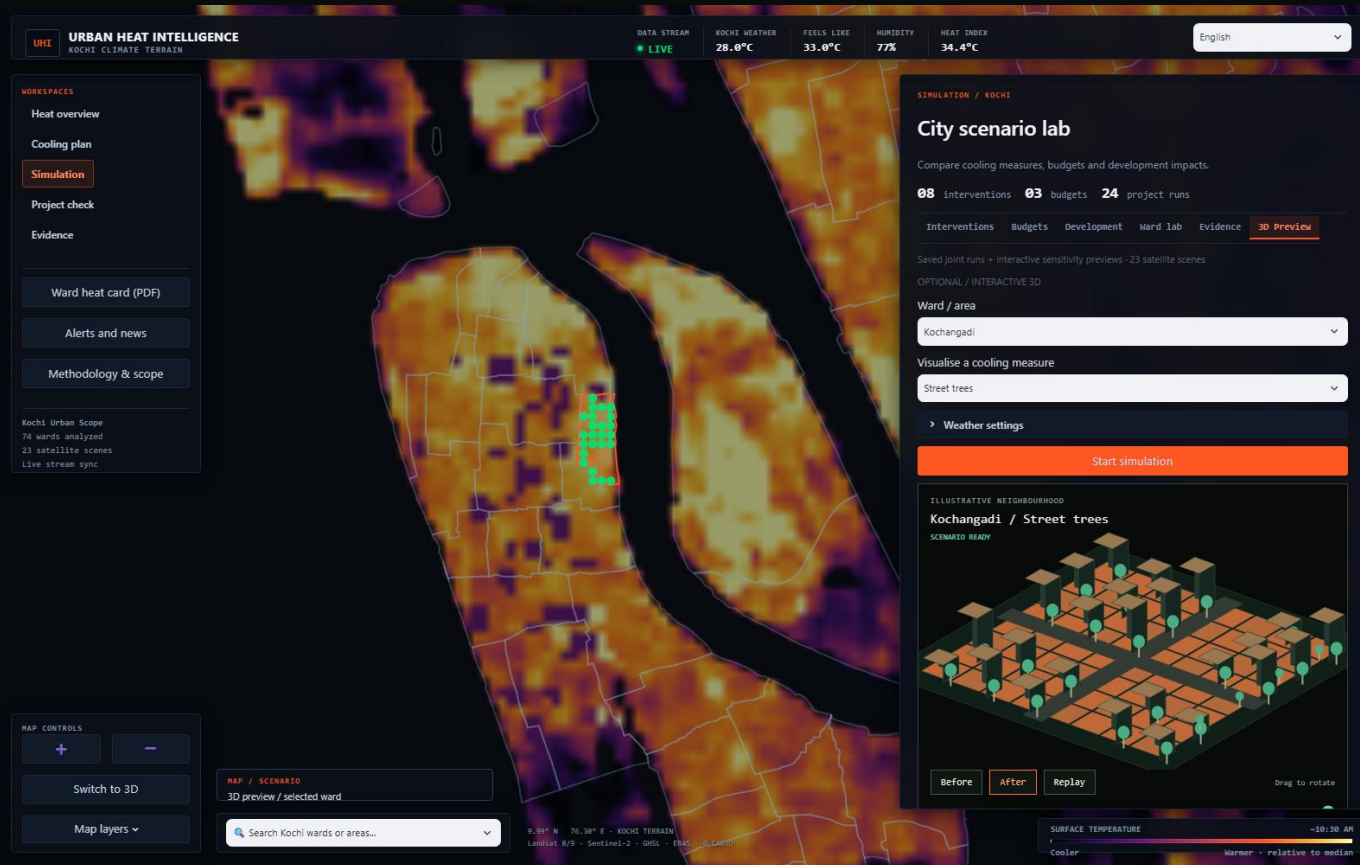
Heat exposure combines surface-heat rank, weather-derived heat index and population; results roll up to wards.

WHY PHYSICS + ML?

ML captures local patterns; explicit formulas cover interventions with weak analog evidence. Spatial validation tests transfer across areas.

One map. Six ways to explore a scenario.

A map-first dashboard with ward selection, English/Malayalam interfaces and downloadable scenario reports.



INTERVENTIONS

8 entries: trees, canal-bank strips, mangroves, ponds, cool/green roofs, cool pavements and zero-credit canals.

BUDGETS + DEVELOPMENT

3 budgets with 2 baseline strategies; 6 project sites × 4 land uses, with available cooling offsets.

WARD LAB + EVIDENCE

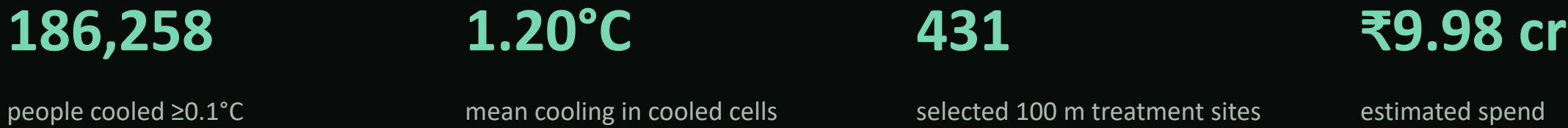
Adjust air temperature, humidity, wind and sunlight; inspect validation, data support and report outputs.

OPTIONAL 3D PREVIEW

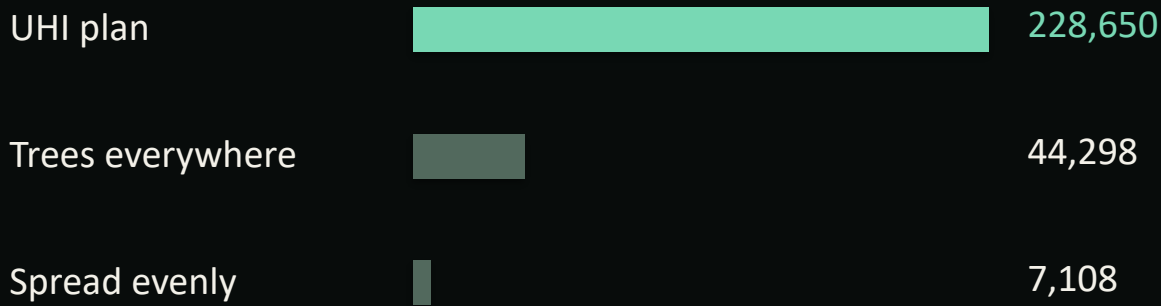
Start simulation, rotate the neighbourhood, replay changes and compare before/after. The scene is illustrative.

₹10 crore: a quantified cooling scenario

Saved estimates for the wider study area, not measured post-construction outcomes.



TOTAL COOLING BENEFIT / person·°C



Mix: 338 street-tree sites · 87 cool roofs · 5 mangrove · 1 canal strip

VALIDATION THAT CAN BE INSPECTED

Spatial R^2 0.829 | RMSE 1.40°C
140 held-out spatial blocks (~2 km)
Linear baseline: R^2 0.616; RMSE 2.10°C

2017–2024 back-test: 8,000 changed cells; MAE 1.40°C ; correlation 0.35. The unconstrained model scores slightly better (R^2 0.839).

Useful today. Explicit about what comes next.

A working planning prototype with traceable outputs, local-language access and honest evidence boundaries.

WHAT IS READY

Interactive heat and ward views; saved cooling plans; 24 development scenarios; weather sensitivity; optional 3D preview; report downloads and standard Malayalam.

DELIVERY + DEMO

Python / Streamlit / XGBoost, serving a frozen data build. Demo: choose a ward → inspect heat → compare a plan → start 3D → download the report.

HOW TO INTERPRET IT

Surface heat is not felt air temperature. City-scale weather is not a ward sensor. Scenarios are indicative; project screening is not planning permission.

PRIORITY NEXT STEPS

Obtain ECOSTRESS afternoon imagery and CPCB station data; produce the matched back-test; field-check costs and outcomes. SOLWEIG / InVEST remain future scope.

THE VALUE: make heat decisions spatial, budget-aware, explainable and reviewable.